USABILITY TESTING & ANALYSIS

Usability Test Report

CampusEats - mobile web food ordering

Test plan, moderated sessions, metrics, findings and design improvements

Prepared by	[Your name / team]
Date	28 September 2026
Method	Moderated, task-based usability testing (5 participants)
Product version	CampusEats web prototype v1.0
Report status	Final draft for review

About this document. CampusEats is a hypothetical product and every participant, score and quote in this report is **illustrative sample data** that shows how a complete usability study is planned, run, analysed and reported. Swap in your own product and session data, keeping the structure, to submit it as your own work.

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1. Executive summary

Five participants completed five realistic tasks on the CampusEats mobile web prototype during 40-minute moderated sessions. The goal was to find out where people struggle when ordering food, how severe each problem is and what to change first.

76%	63/100	34	20%
Tasks completed without help (19 of 25 attempts)	Mean SUS score (industry average is 68)	Total errors observed (17 of them on T3)	Success on T3, applying a promo code (worst task)

Headline findings

- **Promo codes are effectively hidden.** Only 1 of 5 participants applied the code without help. This is a severity 4 (usability catastrophe) issue.
- **The UPI payment hand-off loses users.** One participant returned from the UPI app to a blank screen and abandoned the order (severity 3).
- **Discovery and cart tasks work well.** Search, categories, item customisation and order tracking scored above 80% success and a SEQ of 5.6 or higher.
- **Overall perceived usability is below average.** A mean SUS of 63 sits in the 'marginal' band, pulled down by the checkout experience.

Top recommendations

(1) Show a visible 'Add promo code' field on the cart and payment screens with specific error messages. (2) Add a payment-pending screen with retry and 'check status' actions after the UPI return. (3) Replace the

price slider with preset price chips. Retest with 5 new participants against the targets in Section 10.

2. Introduction and objectives

Usability testing means watching representative users try to complete realistic tasks with a product while thinking aloud. It exposes problems that designers and developers, who know the product too well, cannot see. Testing early and repeatedly makes fixes cheap and turns design into an evidence-based, iterative loop: *test, analyse, redesign, retest*.

2.1 Product under test

CampusEats is a mobile web app that lets students and staff order food from campus canteens. Core flow: browse or search, filter, customise an item, add to cart, apply offers, pay (UPI, card or wallet) and track the order. The tested build is a clickable high-fidelity prototype connected to sample menu data.

2.2 Research objectives

- Measure whether first-time users can complete the five core ordering tasks unaided.
- Identify the specific screens, labels and interactions that cause errors or hesitation.
- Benchmark satisfaction using the System Usability Scale (SUS) so later versions can be compared.
- Produce a prioritised list of design changes and a clear retest plan.

2.3 Research questions

- Can users find and filter meals by price and diet without guidance?
- Do users understand how to apply a discount and complete payment confidently?
- Is the order status clear enough that users know what is happening after they pay?

3. Usability test plan

Element	Decision
Method	Moderated, in-person, task-based testing with concurrent think-aloud. One moderator and one silent note-taker.
Format	Individual sessions of about 40 minutes: briefing 5 min, tasks 25 min, questionnaires and debrief 10 min.
Environment	Quiet meeting room. Participant's own smartphone (so gestures and screen size are familiar) mirrored to a laptop for observation.
Recording	Screen and audio recorded with written consent. Notes captured in a shared observation log (Appendix C).
Pilot	One pilot session with a colleague to check task wording, timing and recording set-up before real sessions.
Schedule	Pilot: Mon. Sessions P1-P5: Tue-Wed. Analysis and report: Thu-Fri.
Success criteria	At least 80% success on every task, mean SEQ of 5.5 or more, SUS of 68 or more.

3.1 Metrics collected

Metric	Type	How it is measured
Task success rate	Performance	Success = completed unaided. Partial = completed with a hint or workaround. Fail = abandoned or wrong result. Reported as full successes divided by attempts.
Time on task	Performance	Seconds from the moderator's 'go' to the participant stating they are done. Reported as mean per task.

Metric	Type	How it is measured
Error count	Performance	Wrong taps, wrong paths and repeated attempts that the note-taker logs against the task.
SEQ (Single Ease Question)	Attitudinal	Asked after each task: 'Overall, how easy was this task?' on a 1 (very difficult) to 7 (very easy) scale.
SUS	Attitudinal	10-item questionnaire after all tasks. Converted to a 0-100 score (Appendix B). 68 is the widely used average.
Qualitative notes	Behavioural	Think-aloud comments, hesitations, facial reactions and moments of confusion, later grouped into themes.

4. Participants and recruitment

Sample size. Five participants per round is a standard choice for qualitative usability testing because it typically reveals roughly 85% of the major problems, and it is more efficient to run several small rounds than one large one. **Recruitment channels.** Notice-board flyers, the student mailing list and personal contacts, screened with the questions below. Each participant received a Rs. 200 food voucher as thanks.

Screener criteria

- Age 18 or above and a regular smartphone user.
- Has ordered food through any app at least once in the last three months.
- Not involved in designing, building or testing CampusEats.
- Mix of ages, devices and food-app experience to avoid a biased sample.

ID	Age	Occupation	Device	Food-app use
P1	22	Undergraduate student	Android	Weekly
P2	27	Software tester	iPhone	Several times a week
P3	45	School teacher	Android	Occasional
P4	31	Marketing executive	iPhone	Weekly
P5	58	Retired bank officer	Android	Occasional

Participants gave informed consent, were told that the product (not they) was being tested and that they could stop at any time.

5. Task scenarios

Each task is written as a realistic scenario without naming buttons or menu labels, so that participants must find their own way. Tasks were presented in a fixed order because each builds on the previous one.

#	Scenario read to the participant	Success criterion
T1	You are vegetarian and have a budget of Rs. 150 for lunch. Find a suitable meal.	Opens a vegetarian item priced at or below Rs. 150.
T2	You want the Paneer Wrap with extra cheese and no onions. Add it to your cart.	Cart shows the item with both customisations.
T3	A friend told you the code **SAVE20** gives a discount. Use it on your order.	Discount visibly applied to the order total.
T4	You are ready to pay. Complete the order using UPI.	Order-confirmed screen is reached.
T5	You changed your mind. Check the status of your order and cancel it.	Status is read correctly and cancellation is confirmed.

6. Results and metrics

6.1 Summary table

Task	Success (S)	Partial (P)	Fail (F)	Mean time	Errors	Mean SEQ
T1 Find a vegetarian meal under Rs. 150	100%	0	0	66 s	5	6.0
T2 Customise an item and add it to cart	100%	0	0	45 s	2	6.4
T3 Apply the promo code SAVE20	20%	2	2	131 s	17	2.6
T4 Check out and pay via UPI	80%	0	1	120 s	7	4.4
T5 Track the order, then cancel it	80%	1	0	40 s	3	5.6
Overall	**76%**	3	3	**80 s**	**34**	**5.0**

Success = completed unaided. Green: at least 80%. Amber: 60-79%. Red: below 60%. Partial completions are not counted as successes. Overall time is the mean of the five task means.

6.2 Task success

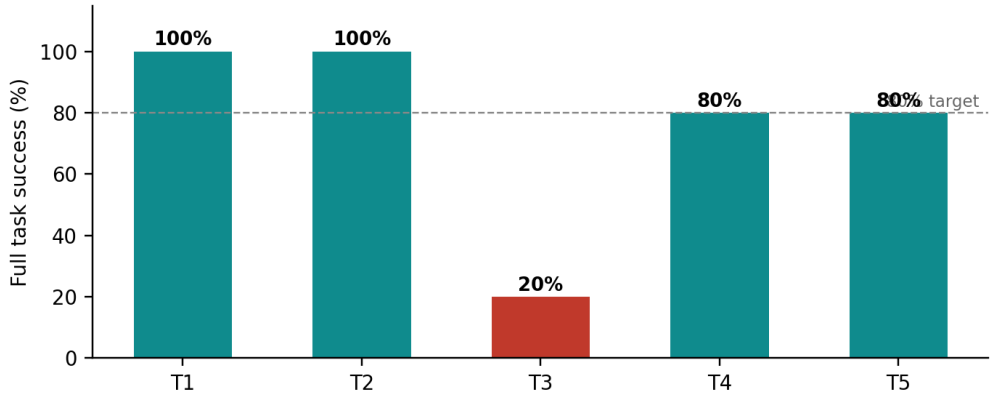


Figure 1 - Task success rate by task. Only T3 falls far below the 80% target.

6.3 Time on task

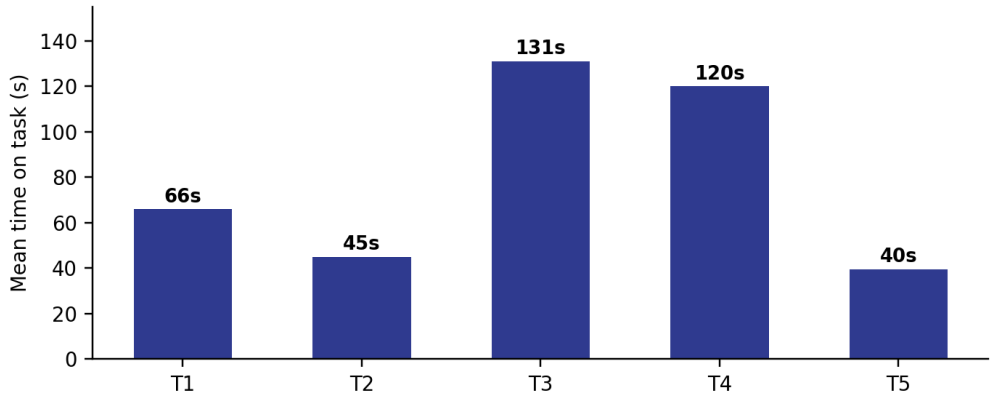


Figure 2 - Mean time on task. T3 and T4 take about three times as long as browsing and cart tasks.

6.4 Perceived usability (SUS)

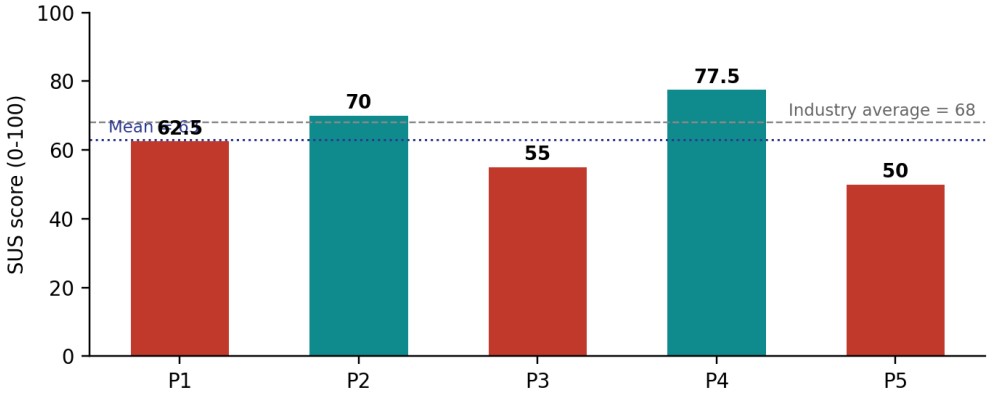


Figure 3 - SUS score per participant. Mean = 63.0. Only P2 and P4 rate the product above the average of 68.

Interpretation. Speed and accuracy on browsing tasks are good, but the two checkout tasks (T3 and T4) account for 24 of the 34 errors and the two lowest SEQ ratings. Satisfaction scores mirror this: participants who struggled with the promo code (P1, P3, P5) gave the lowest SUS scores, while the two who recovered quickly (P2, P4) rated the product above average.

7. Observations and user feedback

Notes from the observation log were affinity-mapped into themes. Representative think-aloud comments are shown below (illustrative).

7.1 Themes

Theme	What was observed	Participants
Hidden promo entry	Participants scrolled the cart and payment screens repeatedly, tapped 'Offers' in the menu, then gave up or asked for a hint.	P1, P2, P4, P5
Unhelpful error text	Entering the code in the wrong field returned 'Invalid input'. People assumed the code itself was wrong.	P1, P4, P5
UPI return confusion	After approving payment in the UPI app, the browser showed a blank page for several seconds with no status message.	P3, P5
Imprecise price filter	The slider was hard to set to exactly Rs. 150 on a small screen and participants tried repeatedly.	P1, P3, P5
Cancel is not discoverable	'Cancel order' sits under a 'Manage' link; one participant tapped the back arrow instead.	P3
Order timeline is liked	The step-by-step status timeline was praised as clear and reassuring.	P1, P2, P4

7.2 Selected quotes

"I know there's a coupon option somewhere, but I honestly can't see where." - P1 (T3)

"It says invalid, so maybe the code is expired? I don't know what I did wrong." - P4 (T3)

"I paid, and now it's just white. Did it go through? I'm scared to press anything." - P5 (T4)

"I like that I can see the order moving from received to preparing to ready." - P2 (T5)

7.3 Post-test debrief (what worked well)

- Clean visual layout and readable food photos (mentioned by 4 of 5).
- Quick, forgiving item customisation (mentioned by 3 of 5).
- Category chips such as 'Breakfast' and 'Snacks' make browsing feel effortless (mentioned by 3 of 5).

8. Findings and severity ratings

Each issue is rated on a 0-4 scale that combines frequency, impact on the task and persistence (whether users learn to work around it). The scale is defined in Appendix D. Heuristics refer to Nielsen's 10 usability heuristics.

ID	Finding	Task	Evidence	Heuristic	Sev.
F1	Promo code entry is hidden behind a small grey 'Have a coupon?' link at the bottom of the payment screen.	T3	4/5 struggled	Visibility of system status; Recognition rather than recall	**4**
F2	Error message 'Invalid input' does not say what is wrong or how to fix it.	T3	3/5	Help users recognise and recover from errors	**3**
F3	No payment-pending or retry screen after returning from the UPI app; blank page for 5-8 seconds.	T4	2/5 (1 fail)	Visibility of system status; Error prevention	**3**
F4	Price filter is a slider with no numeric input, so exact values are hard to set on a small touch screen.	T1	3/5	Flexibility and efficiency of use	**2**
F5	'Cancel order' is hidden under a 'Manage order' link and looks like a secondary text link.	T5	1/5 (partial)	Match between system and real world; Recognition	**2**
F6	'Add to cart' button falls below the fold in the customisation sheet on small screens.	T2	2/5	Aesthetic and minimalist design; Efficiency	**2**
F7	Order tracking timeline is clear and quick to read (positive finding).	T5	3/5 praised	Visibility of system status	+

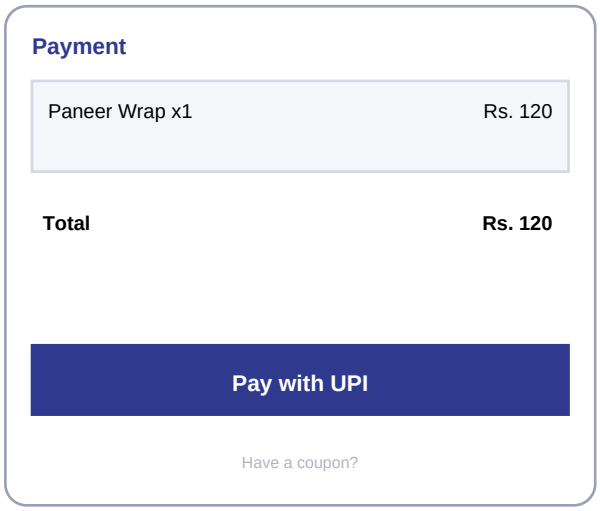
Root-cause view. F1 and F2 together explain why T3 has the lowest success rate, the highest error count and the lowest SEQ. Fixing them is likely to raise SUS more than any other single change because checkout is the moment where frustration matters most.

9. Design improvement recommendations

ID	Recommendation	Fixes	Impact	Effort	Priority
R1	Add a visible 'Add promo code' input with an Apply button on the cart and payment screens. Show the applied discount as a line item.	F1	High	Low	P1
R2	Rewrite errors to be specific and constructive, e.g. 'SAVE20 needs an order of Rs. 199 or more'. Keep the typed code in the field.	F2	High	Low	P1
R3	Add a 'Confirming your payment' screen with a spinner, a 'Check status' button and a retry option when returning from UPI.	F3	High	Medium	P1
R4	Replace the price slider with preset chips (Under 100 / 150 / 200) plus a custom number field.	F4	Medium	Low	P2
R5	Show 'Cancel order' as a clear button on the tracking screen while cancellation is still allowed.	F5	Medium	Low	P2
R6	Make the 'Add to cart' bar sticky at the bottom of the customisation sheet.	F6	Medium	Low	P2

9.1 Example redesign: promo code (R1 and R2)

Before (v1.0): coupon link is tiny and below the pay button



After (v1.1): visible promo field with instant feedback

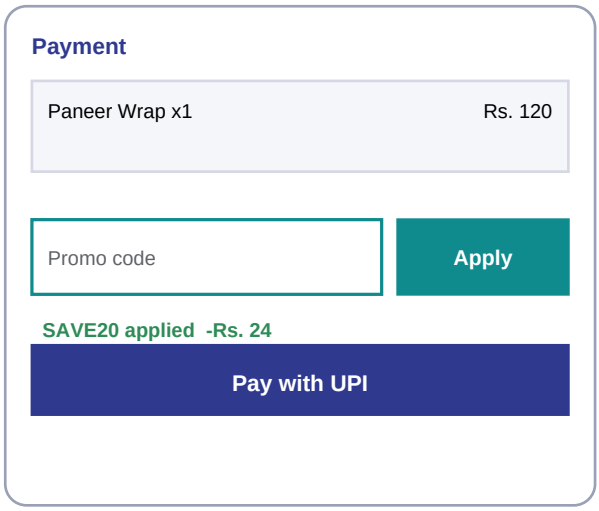


Figure 4 - Low-fidelity wireframes comparing the tested design with the proposed change.

10. Iteration plan and retest criteria

Usability work is iterative. Fixes are implemented in priority order and validated in a second round with new participants, so that improvements can be measured against Round 1.

Step	Activity	Output
1. Test (done)	Round 1 moderated sessions, 5 participants.	This report and baseline metrics.
2. Prioritise	Team workshop to agree on R1-R6 using impact and effort.	Backlog with owners and dates.
3. Redesign	Update wireframes and prototype (v1.1) for P1 and P2 items.	Clickable prototype v1.1.
4. Retest	Round 2 with 5 new participants, same tasks, same metrics.	Comparison of v1.0 against v1.1.
5. Decide	Ship, or iterate again if targets are not met.	Release decision and updated report.

Round 2 targets

Metric	Round 1 (v1.0)	Round 2 target (v1.1)
T3 task success	20%	80% or more
T3 mean time	131 s	60 s or less
T4 task success	80%	100%
Overall task success	76%	90% or more
Total errors	34	15 or fewer
Mean SUS	63	68 or more (stretch: 75)

11. Limitations and conclusion

Limitations

- Five participants reveal major problems but do not provide statistically reliable percentages. Treat success rates as indicative.
- All participants were campus-based, so results may not generalise to other user groups.

- Testing used a prototype with sample data. Real network delays in UPI payments could make F3 better or worse.
- The order of tasks was fixed, so learning effects may slightly favour later tasks.

Conclusion

CampusEats supports discovery and ordering well, but the checkout stage undermines an otherwise easy experience. The promo code (F1, F2) and UPI hand-off (F3) issues are the highest-value fixes and are inexpensive to implement. Round 2 testing against the stated targets will confirm whether the changes improve usability.

Appendix A - Moderator script and consent

A.1 Welcome script (read aloud)

> *"Thank you for coming. Today we are testing the app, not you, so there are no right or wrong answers and you cannot make a mistake. Please think aloud, saying what you are looking for, what you expect and what confuses you. I may not answer questions during the tasks because I want to see how the app works without help, but I will answer them at the end. You can stop at any time. With your permission, we will record the screen and your voice; the recording is used only for analysis and will be deleted after the study."*

A.2 Session flow

Time	Step
0-5 min	Welcome, consent form, explain think-aloud, start recording.
5-8 min	Pre-test questions: age, device, food-app habits, expectations.
8-33 min	Tasks T1-T5. After each task ask the SEQ. Probe neutrally: 'What were you expecting to happen?'
33-40 min	SUS questionnaire, debrief ('What did you like most? What would you change first?'), thank and give voucher.

A.3 Moderator rules

- Stay neutral. Never lead ('Did you see the button on the right?').
- Give a hint only after two minutes of no progress and record the task as Partial.
- Use silence. Wait before prompting. Let users recover on their own.
- Do not defend the design. Note the problem and move on.

A.4 Consent form (summary)

I understand that (1) participation is voluntary and I may withdraw at any time; (2) the session will be screen and audio recorded; (3) my data will be anonymised in reports; (4) recordings are stored securely and deleted after analysis. Name: _____ Signature: _____ Date: _____

Appendix B - Questionnaires

B.1 Single Ease Question (after each task)

"Overall, how difficult or easy was the task to complete?" 1 = Very difficult ... 7 = Very easy.

B.2 System Usability Scale (after the session)

Participants rate each statement from 1 (strongly disagree) to 5 (strongly agree).

#	Statement	Score as
1	I think that I would like to use this system frequently.	Response - 1
2	I found the system unnecessarily complex.	5 - Response
3	I thought the system was easy to use.	Response - 1
4	I think I would need the support of a technical person to use this system.	5 - Response
5	I found the various functions in this system were well integrated.	Response - 1
6	I thought there was too much inconsistency in this system.	5 - Response
7	I would imagine that most people would learn to use this system very quickly.	Response - 1
8	I found the system very cumbersome to use.	5 - Response
9	I felt very confident using the system.	Response - 1
10	I needed to learn a lot of things before I could get going with this system.	5 - Response

Scoring. For odd items subtract 1 from the response, for even items subtract the response from 5. Add the ten values and multiply by 2.5 to get a score from 0 to 100. Example: responses 4,2,4,2,4,2,4,2,4,2 give (3+3+3+3+3+3+3+3+3+3) x 2.5 = 75. Score meaning: below 50 poor, 50-68 marginal, 68-80 good, above 80 excellent.

Appendix C - Raw session data and observation log

C.1 Raw data per participant

Each cell shows result (S = success, P = partial, F = fail), time in seconds and number of errors.

Task	P1	P2	P3	P4	P5
T1	S - 62 s - 1 err	S - 48 s - 0 err	S - 75 s - 2 err	S - 55 s - 0 err	S - 90 s - 2 err
T2	S - 40 s - 0 err	S - 35 s - 0 err	S - 52 s - 1 err	S - 38 s - 0 err	S - 60 s - 1 err
T3	F - 150 s - 4 err	P - 120 s - 3 err	S - 95 s - 2 err	F - 160 s - 5 err	P - 130 s - 3 err
T4	S - 110 s - 1 err	S - 95 s - 1 err	S - 130 s - 2 err	S - 85 s - 0 err	F - 180 s - 3 err
T5	S - 30 s - 0 err	S - 25 s - 0 err	P - 70 s - 2 err	S - 28 s - 0 err	S - 45 s - 1 err

Participant	P1	P2	P3	P4	P5	Mean
SUS score	62.5	70	55	77.5	50	63.0

C.2 Observation log template

Copy this table for each session. Record facts (what the user did and said) separately from interpretations.

Time	Task	Observation (what happened / said)	Error?	Severity

Appendix D - Severity scale and publishing on GitHub

Rating	Label	Meaning
0	Not a problem	Disagreement; not a usability problem.
1	Cosmetic	Fix only if time is available.
2	Minor	Low priority. Slows users slightly.
3	Major	High priority. Causes significant difficulty or failure for some users.
4	Catastrophe	Must fix before release. Blocks users from completing a key task.

Suggested repository structure

Path	Content
/report/Usability_Testing_and_Analysis_Report.pdf	This report.
/data/session_data.csv	Raw success, time and error data per participant and task.
/materials/	Consent form, moderator script, task cards, SUS questionnaire.
/design/	Screenshots or Figma exports: before (v1.0) and after (v1.1).
/README.md	Project summary, method, key results and links.

References. Nielsen, J. (1994) 10 Usability Heuristics for User Interface Design. Brooke, J. (1996) SUS - A quick and dirty usability scale. Nielsen, J. and Landauer, T. (1993) A mathematical model of the finding of usability problems. Sauro, J. and Lewis, J. (2016) Quantifying the User Experience.